

Fergal Riordan

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PROFILE

ML Engineer with **First-Class Honours MEng/BEng degrees in Electronic & Computer Engineering** from Trinity College Dublin. Professional background includes **1 year of deploying production-grade AI systems** and ~2 years of freelance experience in RLHF-based model alignment. Specialist in generative modeling and agentic orchestration, with a strong foundation in signal processing and mathematical optimisation.

TECHNICAL SKILLS

Languages:	Python, C++, SQL, C
ML Frameworks:	PyTorch, Hugging Face, Scikit-learn, TorchVision, OpenCV
Data & Ops:	Pandas, NumPy, Docker, Git, Azure
Foundations:	Linear Algebra, Probability & Statistics, Optimisation, Signal Processing
Competencies:	Deep Learning Research, Computer Vision, Generative Modeling, Agentic Systems

EXPERIENCE

AI Engineer <i>Channelscaler</i> <i>Python, Microsoft Azure, Cosmos DB, Agentic AI</i> Overview of Channelscaler AI Features	May 2025 – Present Galway, Ireland
- Co-engineered the platform's first AI agent application from scratch, designing extensible orchestration logic for autonomous tool-calling, SQL querying, and RAG. - Fine-tuned Azure Document Intelligence models for invoice auditing, increasing accuracy via iterative evaluation.	
AI Data Annotation Specialist <i>Data Annotation Tech</i> <i>Python, RLHF, Prompt Engineering, Model Evaluation</i>	Sep. 2023 – May 2025 Remote, Ireland
Contributed as a freelancer to a domain-expert team for RLHF-based training of IDE-integrated coding assistants.	

TECHNICAL RESEARCH & PROJECTS

Master's Thesis: Enhancing CycleGAN for Image Translation View Project <i>Python, PyTorch, GANs, Computer Vision, Loss Function Optimisation, Transfer Learning</i>	Sep. 2023 – May 2024
- Proposed a custom architecture with shared generators and a novel timestamp conditioning strategy. - Achieved an improvement of 20% on the Kernel Inception Distance metric over the baseline CycleGAN model.	
Time-Series Analysis: Urban Mobility Forecasting <i>Python, Scikit-learn, Lasso Regression, Statistical Analysis</i>	Oct. 2023 – Dec. 2023
- Forecasted DublinBikes demand using Lasso Regression for feature selection and temporal data analysis. - Evaluated model robustness and classifier performance through k-fold cross-validation and ROC analysis.	
Classical Computer Vision: Draughts Game State Analysis View Project <i>C++, OpenCV, Computer Vision</i>	Sep. 2022 – Nov. 2022
- Built a detection engine using Gaussian Mixture Models (GMM) for motion detection and board-state validation. - Implemented histogram back-projection and probability thresholding for pixel classification and piece segmentation.	

EDUCATION

Trinity College Dublin <i>Master of Engineering & Bachelor of Engineering in Electronic & Computer Engineering</i>	Dublin, Ireland Sep. 2019 – May 2024
- MEng: First-Class Honours (1.1), 80% - BEng: First-Class Honours (1.1), 72% - Erasmus semester at the University of Iceland, Reykjavik	
Christian Brothers College <i>Leaving Certificate</i>	Cork, Ireland Sep. 2013 – Jun. 2019
Ranked in top 70 students nationally - 625/625 points (7 H1 grades)	